Lin–Log Method

Principle

The Lin–Log method assumes that the maximum cyclic stress varies linearly with the logarithm of the number of cycles to failure.

$$\sigma_{max} = a + b\log_{10}(N)$$

The S–N curve is therefore represented by a straight line on a semi-logarithmic plot (Stress: linear scale, Cycles: logarithmic scale).

Example

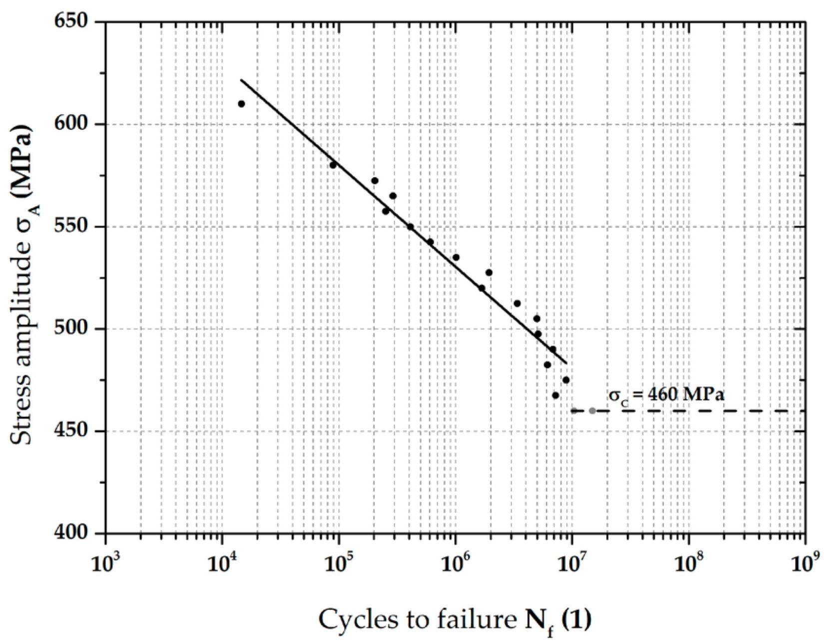

Figure 1 : Linear S–N curve obtained with the Lin–Log method

Curve Identification

1) Experimental fatigue data are collected as pairs

$$(N_i, \ \sigma_{max,i})$$

2) The number of cycles is transformed:

$$x_i = \log_{10}(N_i)$$

3) A linear regression is performed on

$$\sigma_{A,i} = a + bx_i$$

4) The fitted coefficients a(intercept) and b(slope) define the S–N curve.

The coefficients are obtained by least-squares minimization of the vertical residuals in stress.

Assumptions & Recommended use

The method assumes an approximately linear behaviour in semi-log space over the considered cycle range.
It does not model endurance limits or probabilistic scatter explicitly.

Recommended for preliminary analysis and moderate cycle ranges.
Less suitable for strongly curved composite S–N responses or very high cycle fatigue.

Log–Log Method

Principle

The Log–Log method assumes a power-law relationship between stress and fatigue life:

$$\sigma_a = A\,N^{\,b}$$

Taking the logarithm of both sides:

$$\log(\sigma_a) = \log(A) + b\log(N)$$

The S–N curve is therefore represented by a straight line on a log–log plot (Stress: logarithmic scale, Cycles: logarithmic scale).

Example

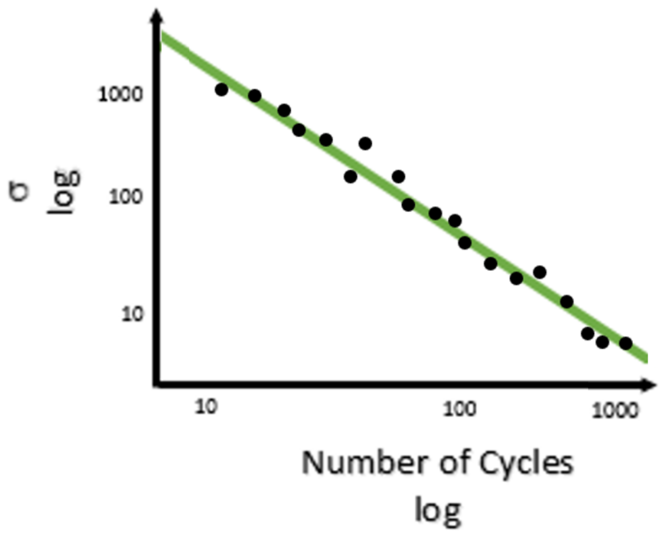

Figure 2 : Power-law S–N curve obtained with the Log–Log method

Curve Identification

1) Experimental fatigue data are collected as pairs

$$(N_i,\ \sigma_{a,i})$$

2) Both variables are transformed:

$$x_i = \log_{10}(N_i), y_i = \log_{10}(\sigma_{a,i})$$

3) A linear regression is performed on

$$y_i = \log(A) + bx_i$$

4) The fitted parameters A and b define the S–N curve:

$$\sigma_a = AN^{\,b}$$

The program determines A and b by least-squares minimization of the vertical residuals in log-stress.

Assumptions & Recommended use

The method assumes fatigue behaviour follows a power-law over the considered cycle range. It naturally captures scale effects and continuous slope evolution.

Widely used for metallic fatigue and composite materials without a clear endurance limit. Less suitable when a fatigue plateau or strong curvature is observed.

Sendeckyj Method

Principle

The Sendeckyj method models fatigue by introducing an equivalent static strength and assuming that fatigue data can be transformed into a common strength distribution.

The deterministic wearout model is written as:

$$\sigma_r = \sigma_e \left[\left(\frac{\sigma_a}{\sigma_e} \right)^{1/s} + (n-1)C \right]^s$$

σ_a: maximum applied cyclic stress

σ_r: residual strength

σ_e: equivalent static strength

n: number of cycles

s: asymptotic slope parameter

C: parameter controlling the flat region at low cycles

For fatigue failure ($\sigma_r = \sigma_a$, $n = N$), the S–N relation becomes:

$$\sigma_e = \sigma_a (1 - C + CN)^s$$

The method then searches the parameters that give the least scatter in the transformed equivalent static strengths, assumed to follow a two-parameter Weibull distribution.

Example

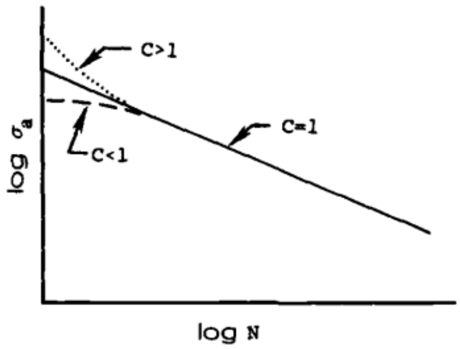

Figure 3 : Effect of parameter C on the S–N curve

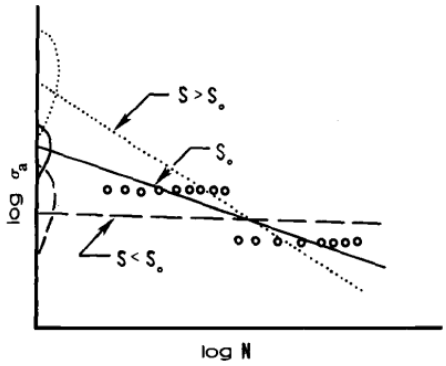

Figure 4 : Effect of parameter s on the S–N curve slope

Curve Identification

1) Experimental fatigue data are collected as:

$$(\sigma_{a,i},\ n_i,\ \sigma_{r,i})$$

or, for failure data:

$$(\sigma_{a,i},\ N_i)$$

2) For assumed values of s and C, each datum is transformed into an equivalent static strength $\sigma_{e,i}$.

3) A two-parameter Weibull distribution is fitted to the σ_e values:

$$P(\sigma_e) = \exp\left[-\left(\frac{\sigma_e}{\beta}\right)^{\alpha}\right]$$

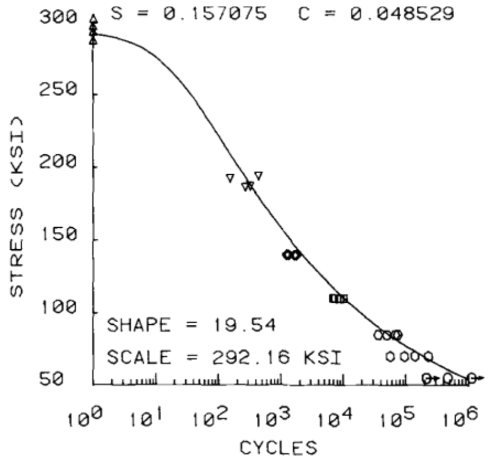

Figure 5 : Example of S–N curve fitted using the Sendeckyj method

4) The optimal fatigue curve is obtained by selecting the parameters s and C
that **maximize the Weibull shape parameter** α, i.e. minimize scatter in the equivalent static strength distribution.

The fitted parameters s, C, α and β define the S–N behaviour.

Assumptions & Recommended use

The method assumes that fatigue behaviour can be represented by a deterministic wearout law and that the transformed equivalent static strengths follow a two-parameter Weibull distribution.

It can incorporate runouts and tab failures through censoring.
It is well suited to composite materials, especially when the S–N response is curved and not well represented by a simple linear or power-law fit.

More advanced than Lin–Log and Log–Log methods, it is appropriate when a probabilistic description of scatter is needed.

Whitney Method

Principle

The Whitney method provides a simplified alternative to wear-out based approaches such as Sendeckyj.

It assumes that fatigue behaviour can be described by:

- a **power-law S–N relationship**,
- a **two-parameter Weibull distribution** for fatigue life.

Unlike Sendeckyj, the degradation process is not explicitly modeled through a residual strength variable. Instead, fatigue variability is described statistically.

The S–N relationship at a given survival probability is written as:

$$\sigma_{\max} = \sigma_0[-\ln(P_s)]^{\frac{p}{\alpha_f}} N^{-p}$$

where:

- $\sigma_{\max}$: maximum cyclic stress
- N: number of cycles to failure
- P_s: survival probability
- α_f: Weibull shape parameter
- p: slope parameter
- σ_0: material parameter

Example

Curve Identification

The identification procedure follows a simplified statistical framework:

1. Fatigue data are grouped by stress level.
2. For each level, a characteristic fatigue life is estimated.
3. The fatigue lives are normalized with respect to their characteristic values.
4. The normalized data are then pooled across all stress levels to identify a single shape parameter α_f, describing the global dispersion.
5. A power-law regression is performed to relate stress and fatigue life.

6. The final S–N curve is obtained for the selected survival probability.

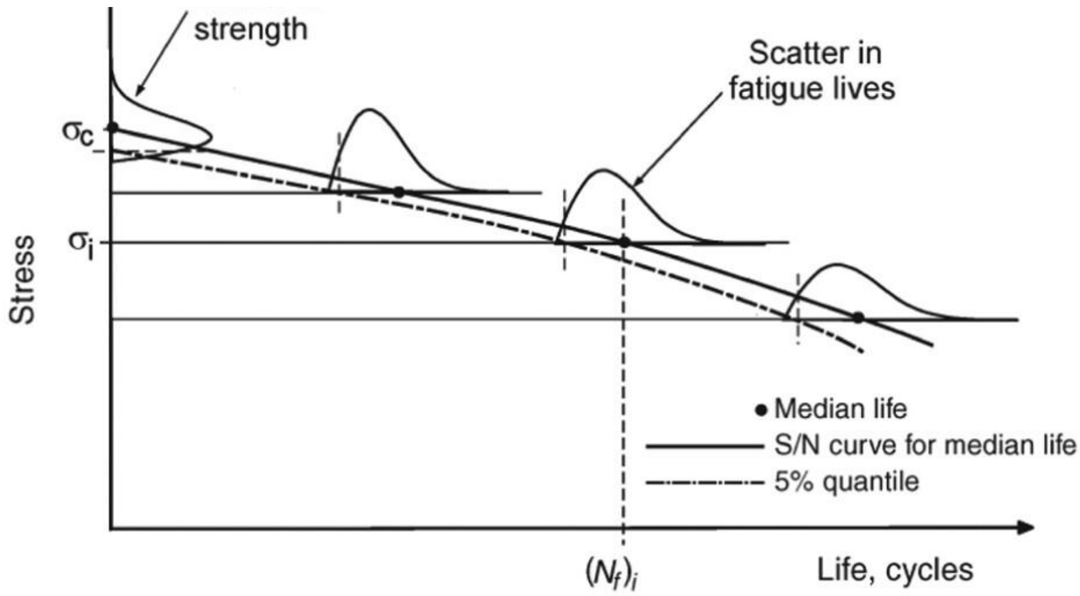

Figure 6 : Illustration of fatigue life dispersion across stress levels and construction of probabilistic S–N curves in Whitney's method, highlighting median behavior and lower reliability quantiles.

Assumptions & Recommended use

The method assumes a consistent statistical behaviour of fatigue life across stress levels and a valid power-law relationship between stress and cycles. It also assumes that fatigue variability can be described globally using a single parameter.

Simpler and more direct than Sendeckyj, the method is well suited for generating probabilistic S–N curves and for quick engineering estimations.

However, it does not explicitly model fatigue damage evolution and may be less effective for complex material behaviour.

Recommended for applications where simplicity and robustness are preferred over detailed physical modeling.

Sources

- **Sendeckyj, G. P. (1981)**
  Fitting Models to Composite Materials Fatigue Data.
  ASTM STP 734, American Society for Testing and Materials, pp. 245-260.
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